

AWS EC2 and EBS Volume Attachment Report

Step-by-step practical report for launching an EC2 Ubuntu instance, creating an Amazon EBS volume, attaching it to EC2, and verifying it through Linux terminal commands.

AWS Region	Asia Pacific (Mumbai) - ap-south-1
Instance Name	Ebs-connect
Instance ID	i-0b81e1e5bf5e41eb5
Operating System	Ubuntu 26.04 LTS
EBS Volume	vol-019e9869a679c8513, 5 GiB, gp3
Device / Mount Point	/dev/nvme1n1 mounted on /mnt/mydata

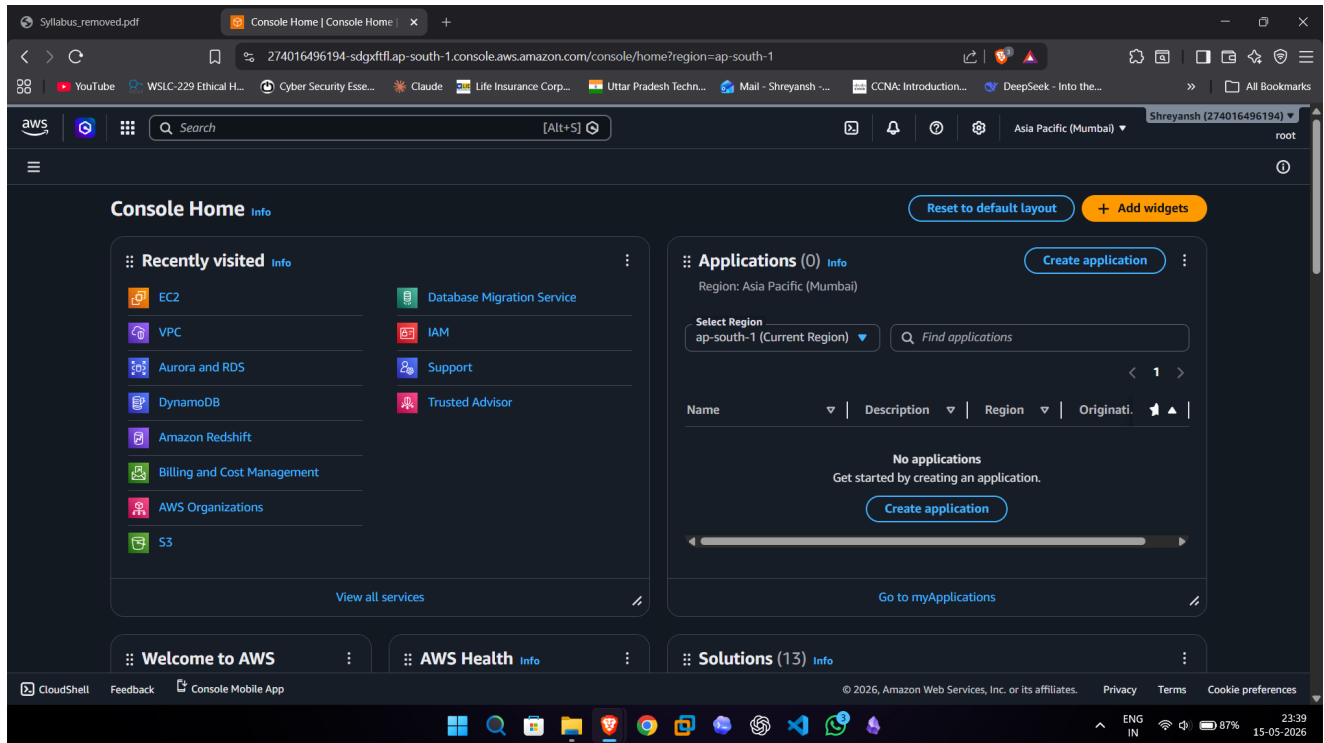
Practical Overview

- Launch an EC2 Ubuntu instance in AWS.
- Create a new key pair for SSH login.
- Configure security group rules for SSH, HTTP, and HTTPS.
- Create a 5 GiB gp3 Amazon EBS volume in the same availability zone.
- Attach the EBS volume to the running EC2 instance.
- Format, mount, and verify the EBS volume from Ubuntu Linux.

Implementation Steps

Step 1: Open AWS Console Home

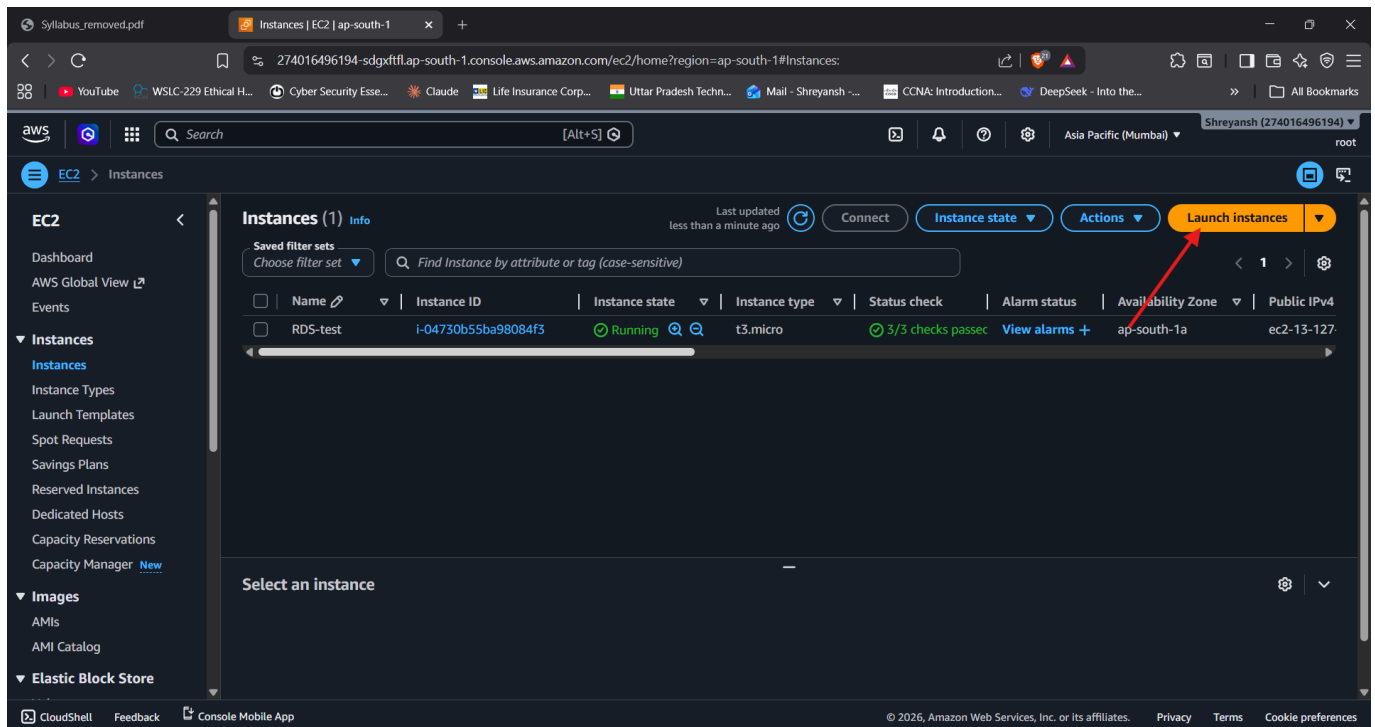
The AWS Console Home page was opened in the Asia Pacific (Mumbai) region.



Screenshot 2026-05-15 233941.png

Step 2: Open EC2 Instances Dashboard

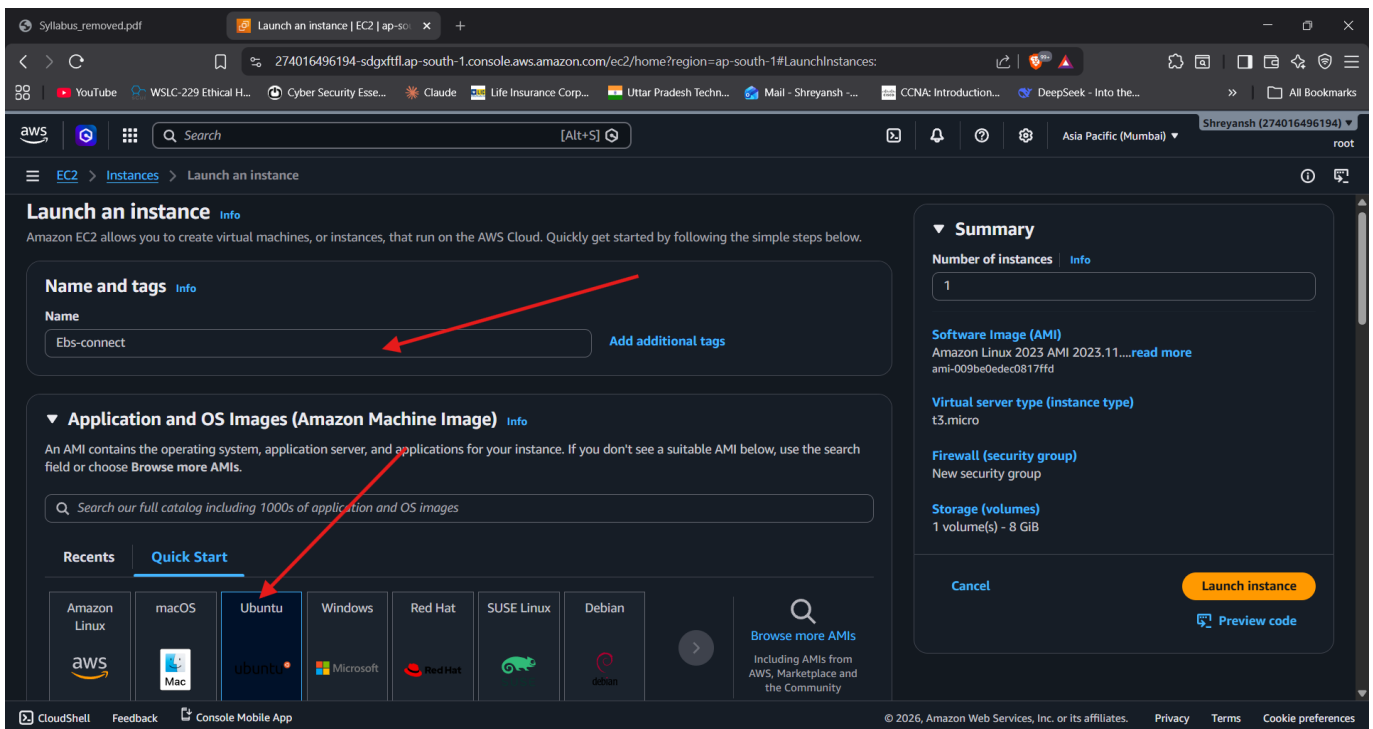
The EC2 Instances page shows the existing running instance and the Launch instances option.



Screenshot 2026-05-15 234022.png

Step 3: Configure Instance Name and AMI

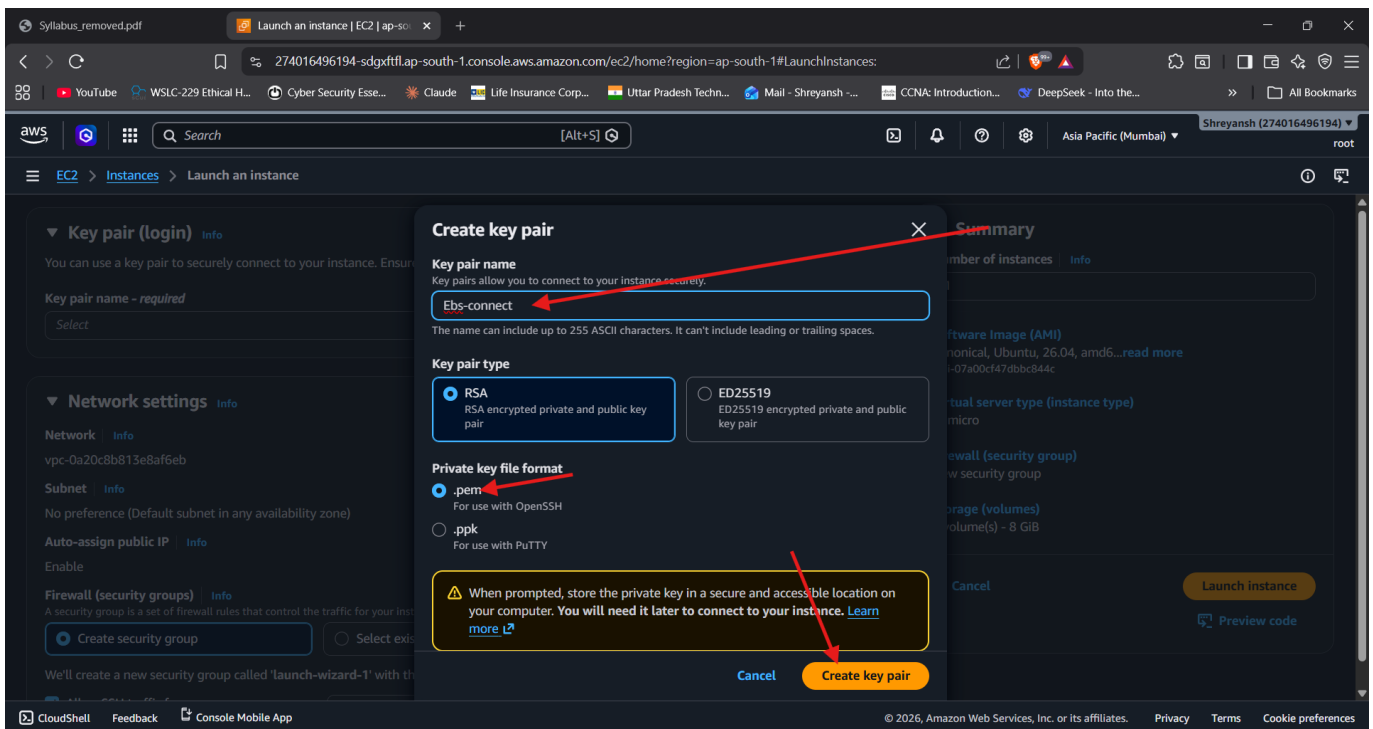
A new EC2 instance named Ebs-connect was created using an Ubuntu AMI and t3.micro instance type.



Screenshot 2026-05-15 234121.png

Step 4: Create Key Pair

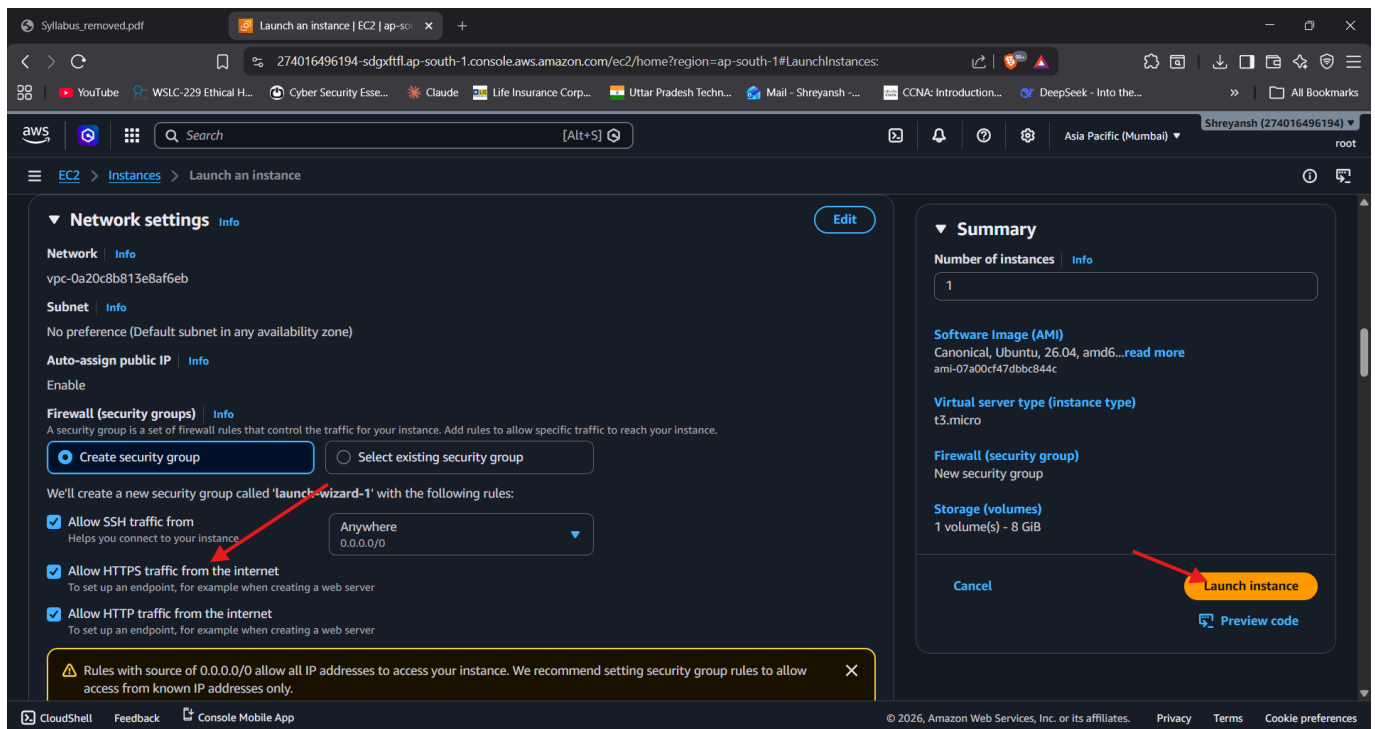
A new key pair named Ebs-connect was created using RSA and .pem format for SSH access.



Screenshot 2026-05-15 234307.png

Step 5: Configure Network and Launch Instance

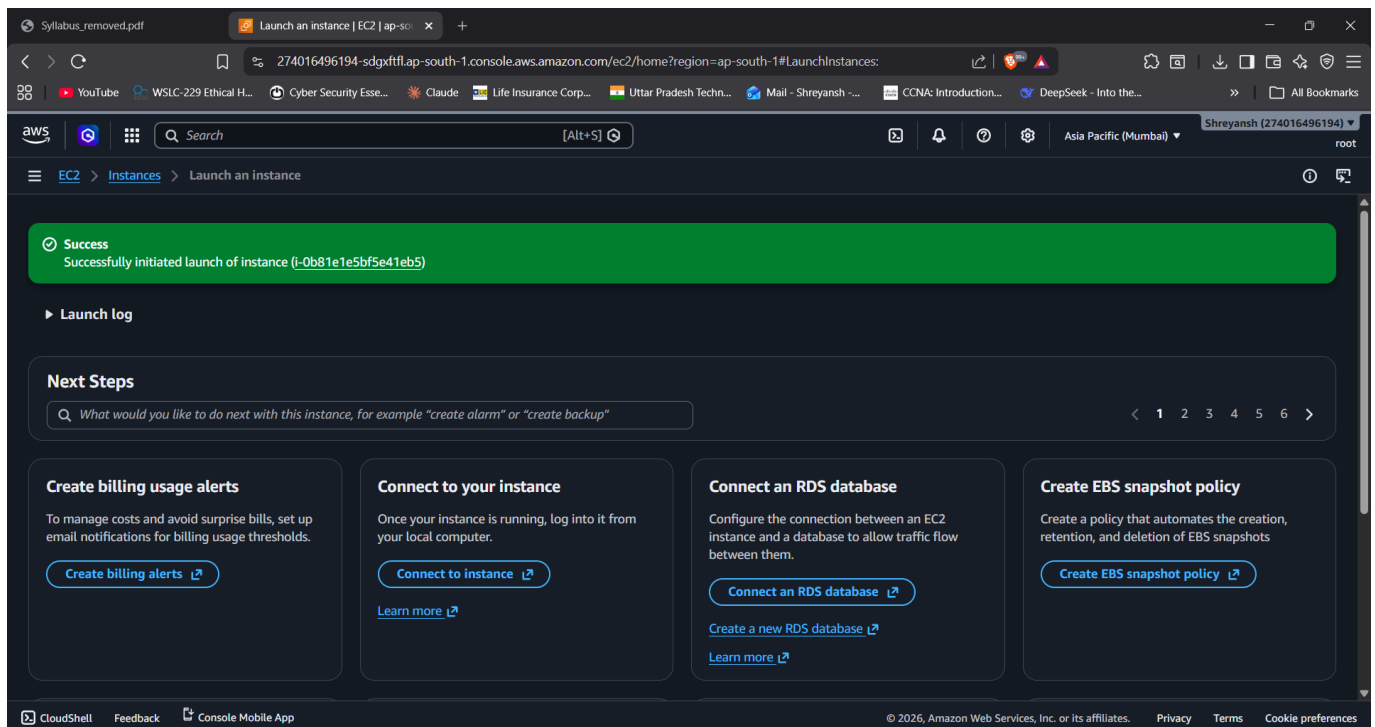
SSH, HTTP, and HTTPS rules were enabled, and the instance launch was started.



Screenshot 2026-05-15 234535.png

Step 6: Verify Successful Instance Launch

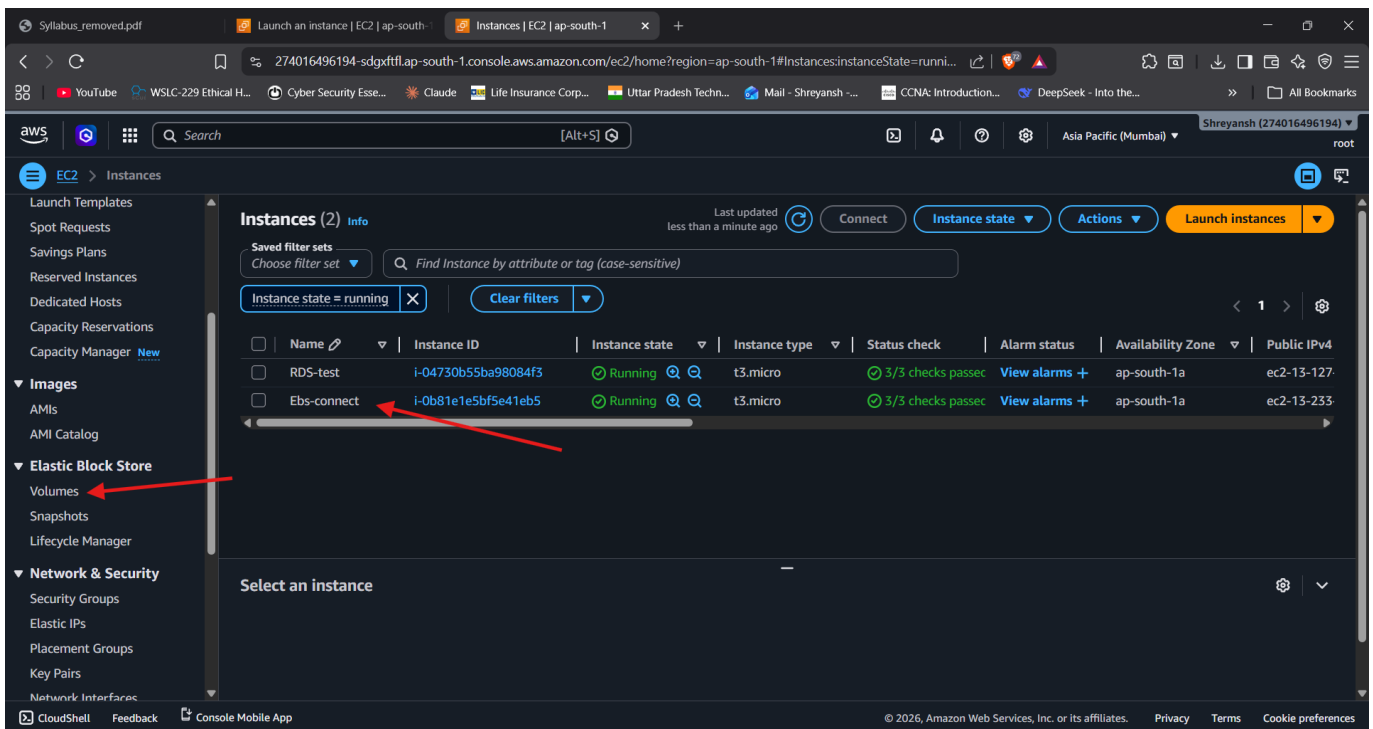
AWS confirmed that the instance i-0b81e1e5bf5e41eb5 launched successfully.



Screenshot 2026-05-15 234646.png

Step 7: Open Elastic Block Store Volumes

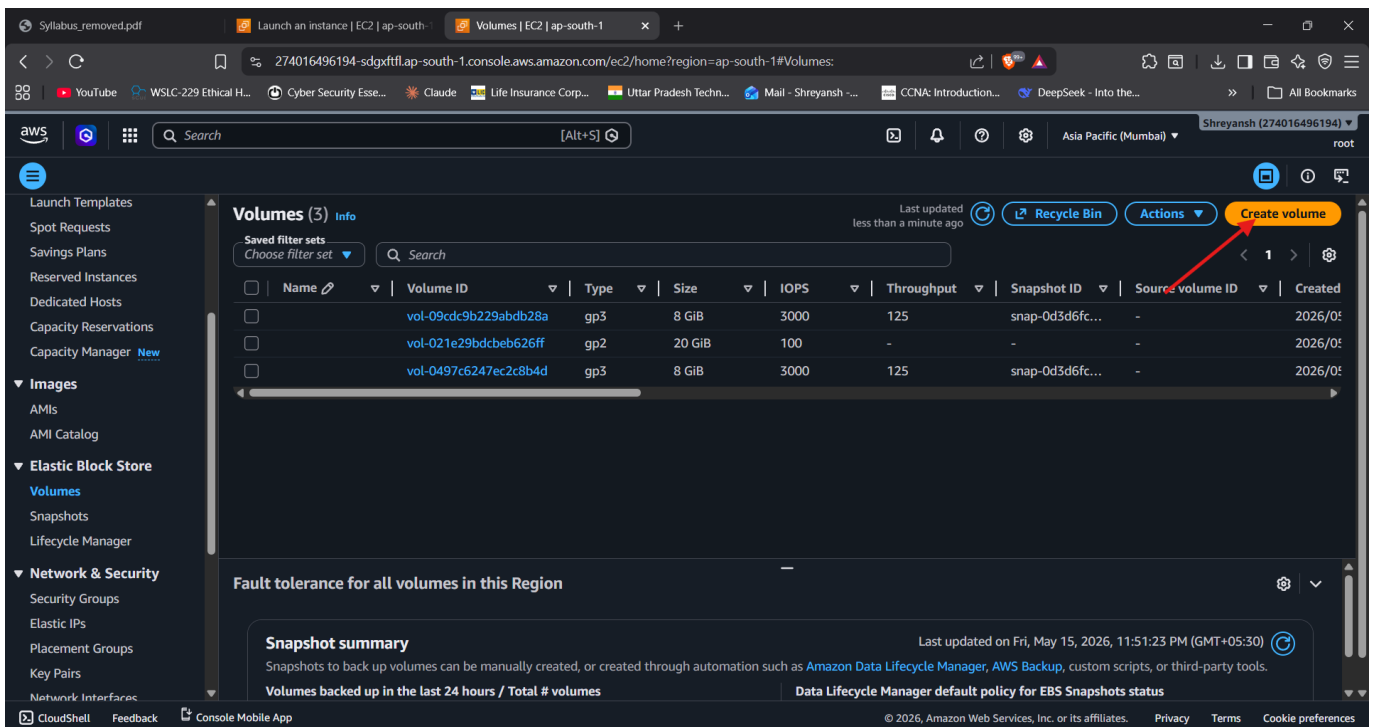
The Elastic Block Store Volumes section was opened to create an additional EBS volume.



Screenshot 2026-05-15 235024.png

Step 8: Create New EBS Volume

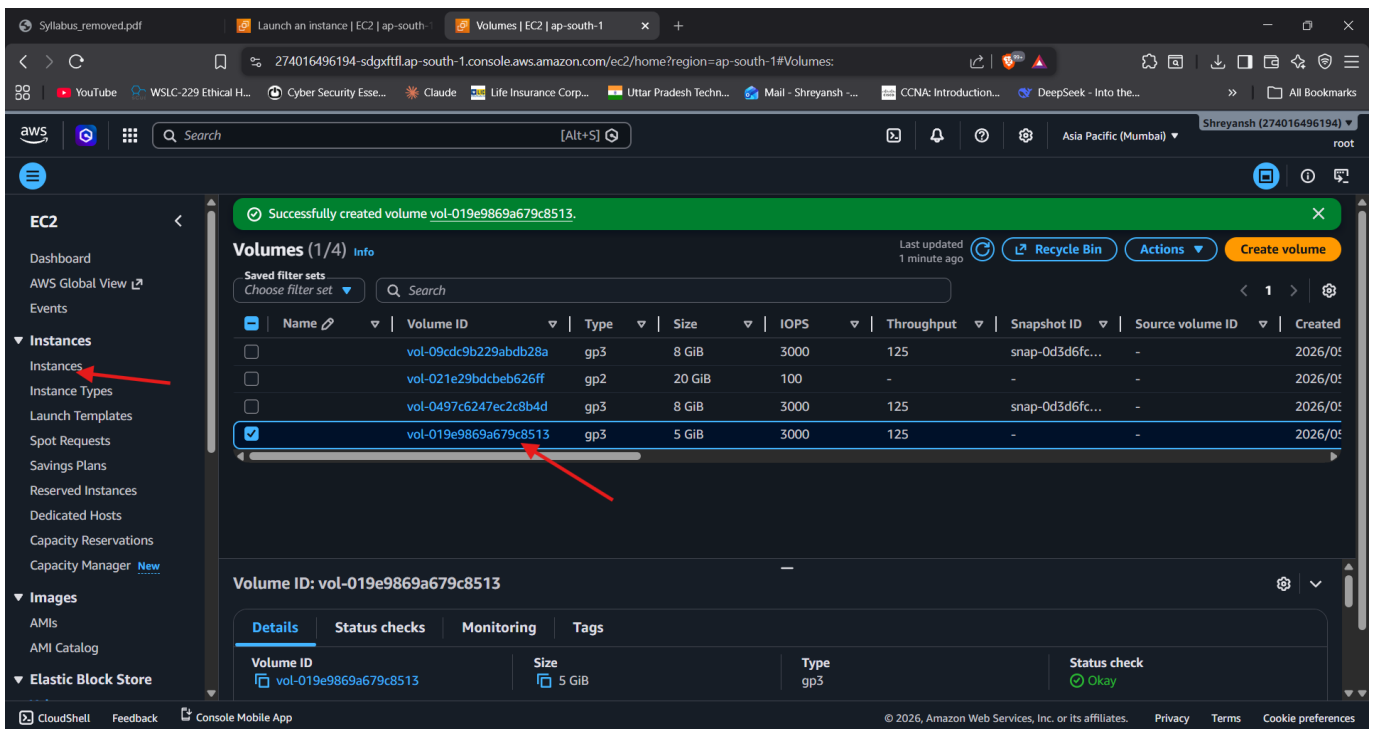
A General Purpose SSD (gp3) volume of 5 GiB was configured in availability zone ap-south-1a.



Screenshot 2026-05-15 235153.png

Step 9: Confirm Volume Creation

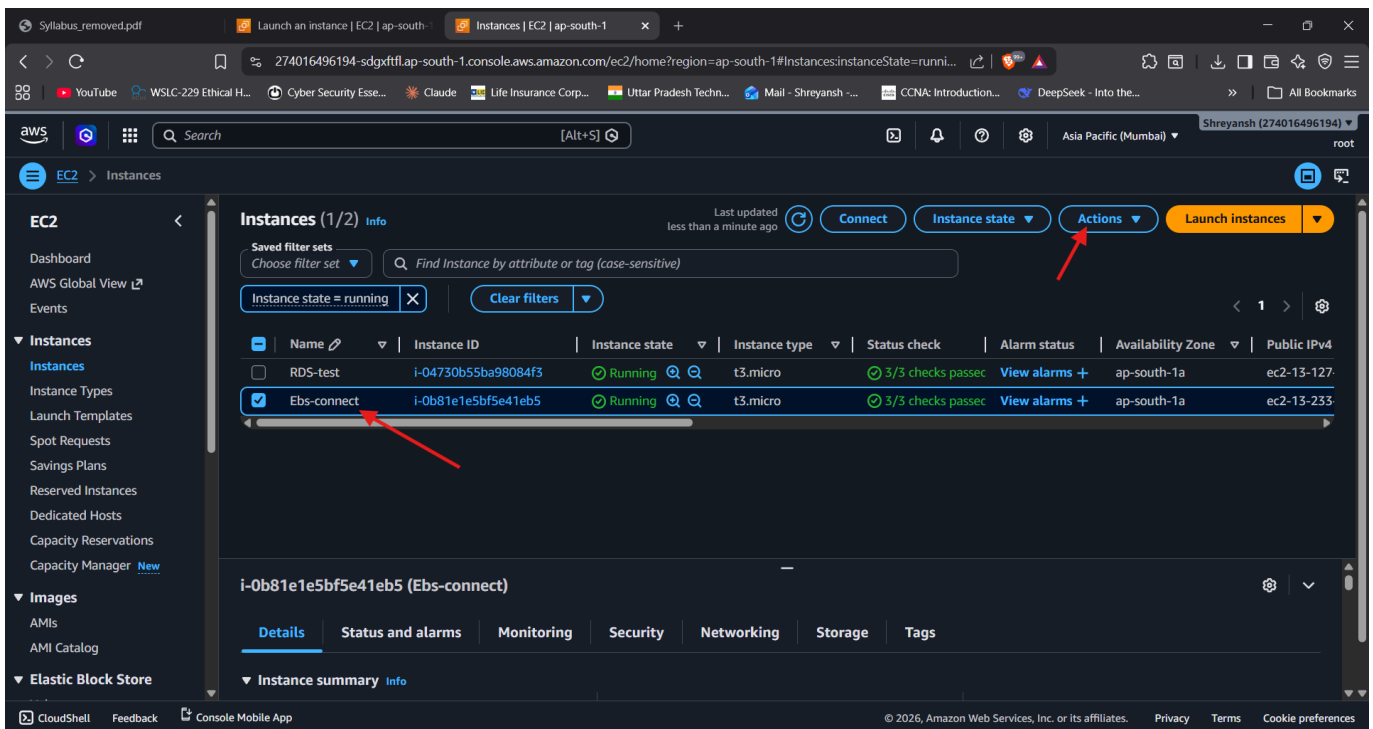
The new volume vol-019e9869a679c8513 was created successfully with 5 GiB size.



Screenshot 2026-05-15 235531.png

Step 10: Select EC2 Instance for Attachment

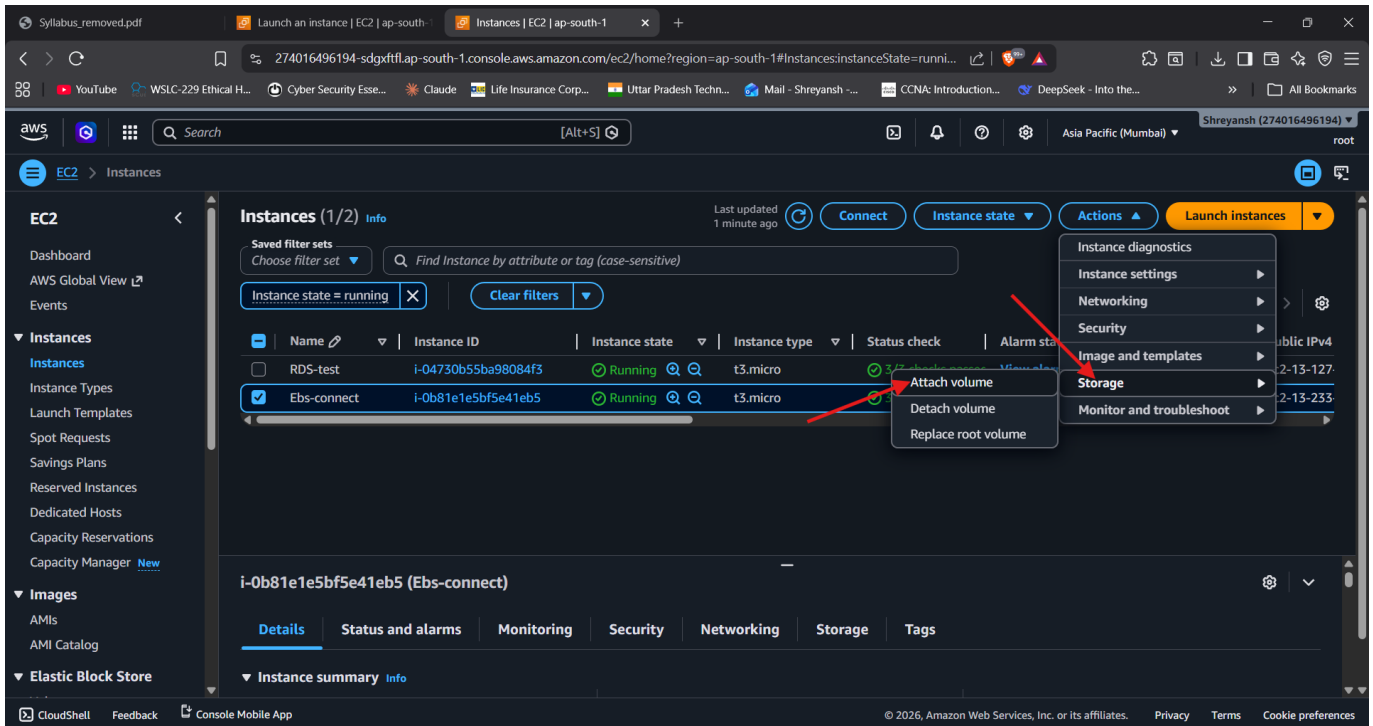
The Ebs-connect instance was selected, and the Storage > Attach volume action was used.



Screenshot 2026-05-15 235639.png

Step 11: Attach EBS Volume

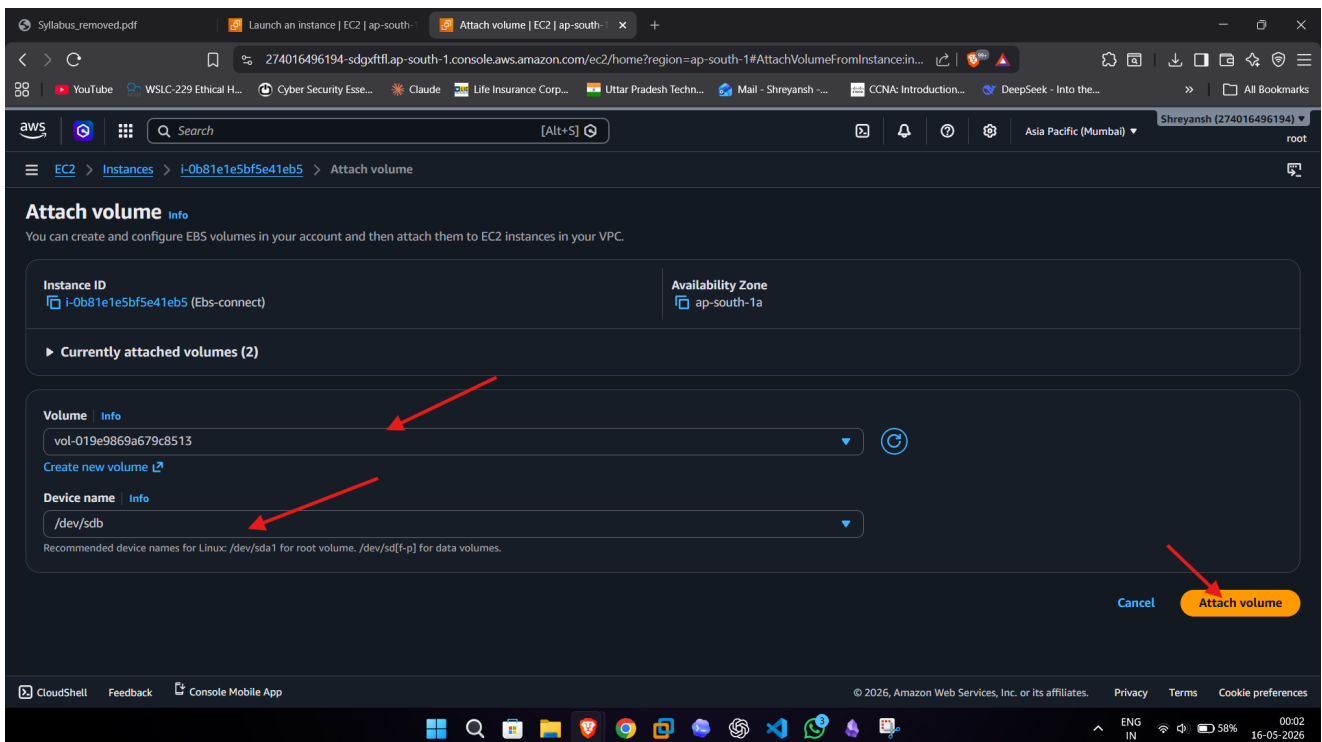
The EBS volume vol-019e9869a679c8513 was attached to the EC2 instance using a Linux device name.



Screenshot 2026-05-15 235715.png

Step 12: Verify Volume in AWS Storage Tab

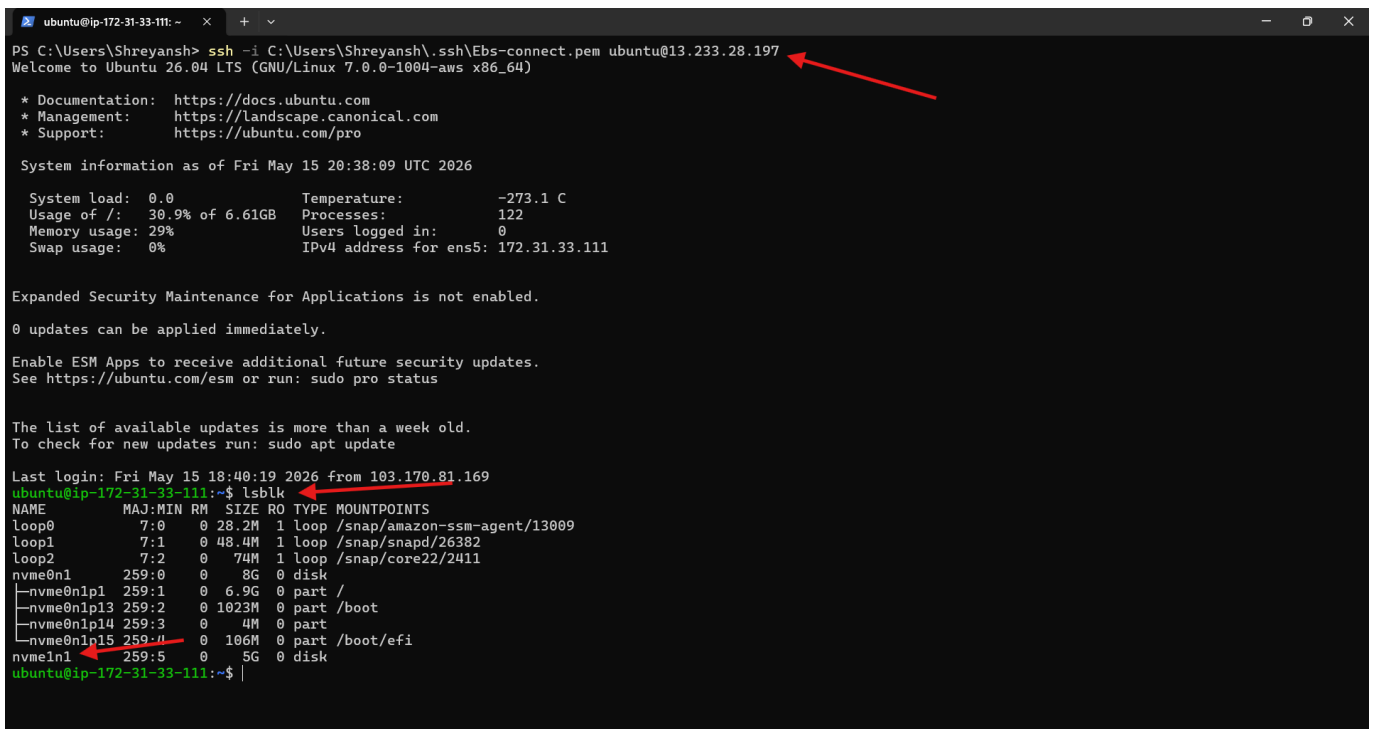
The Storage tab shows both the root volume and the newly attached 5 GiB EBS volume.



Screenshot 2026-05-16 000208.png

Step 13: Connect to EC2 and Check Disk

SSH connection was established from Windows Terminal and lsblk confirmed the new 5 GiB disk.

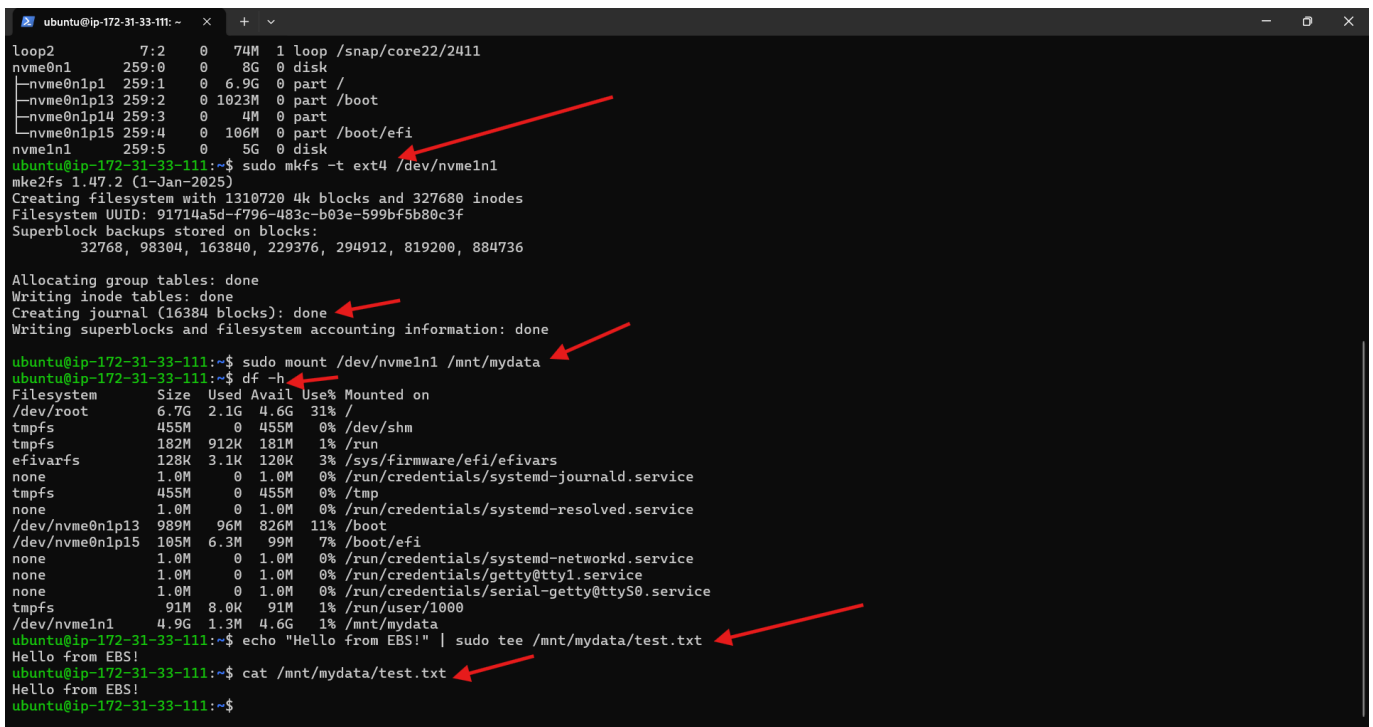


```
ubuntu@ip-172-31-33-111: ~  
PS C:\Users\Shreyansh> ssh -i C:\Users\Shreyansh\.ssh\Ebs-connect.pem ubuntu@13.233.28.197  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Fri May 15 20:38:09 UTC 2026  
  
System load: 0.0      Temperature: -273.1 C  
Usage of /: 30.9% of 6.61GB  Processes: 122  
Memory usage: 29%      Users logged in: 0  
Swap usage: 0%         IPv4 address for ens5: 172.31.33.111  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Fri May 15 18:40:19 2026 from 103.170.81.169  
ubuntu@ip-172-31-33-111:~$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
loop0        7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009  
loop1        7:1    0 48.4M  1 loop /snap/snapd/26382  
loop2        7:2    0 74M   1 loop /snap/core22/2411  
nvme0n1      259:0    0   8G   0 disk  
└─nvme0n1p1  259:1    0  6.9G   0 part /  
└─nvme0n1p13 259:2    0 1023M   0 part /boot  
└─nvme0n1p14 259:3    0   4M   0 part  
└─nvme0n1p15 259:4    0 106M   0 part /boot/efi  
nvme1n1      259:5    0   5G   0 disk  
ubuntu@ip-172-31-33-111:~$
```

Screenshot 2026-05-16 020831.png

Step 14: Format, Mount, and Test EBS Volume

The EBS volume was formatted as ext4, mounted at /mnt/mydata, and tested by writing and reading a file.



```
ubuntu@ip-172-31-33-111: ~  
loop2        7:2    0 74M   1 loop /snap/core22/2411  
nvme0n1      259:0    0   8G   0 disk  
└─nvme0n1p1  259:1    0  6.9G   0 part /  
└─nvme0n1p13 259:2    0 1023M   0 part /boot  
└─nvme0n1p14 259:3    0   4M   0 part  
└─nvme0n1p15 259:4    0 106M   0 part /boot/efi  
nvme1n1      259:5    0   5G   0 disk  
ubuntu@ip-172-31-33-111:~$ sudo mkfs -t ext4 /dev/nvme1n1  
mke2fs 1.47.2 (1-Jan-2025)  
Creating filesystem with 1310720 4k blocks and 327680 inodes  
Filesystem UUID: 91714a5d-f796-483c-b03e-599bf5b80c3f  
Superblock backups stored on blocks:  
32768, 98304, 163840, 229376, 294912, 819200, 884736  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (16384 blocks): done  
Writing superblocks and filesystem accounting information: done  
  
ubuntu@ip-172-31-33-111:~$ sudo mount /dev/nvme1n1 /mnt/mydata  
ubuntu@ip-172-31-33-111:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root       6.7G  2.1G  4.6G  31% /  
tmpfs           455M   0  455M   0% /dev/shm  
tmpfs           182M   0  181M   1% /run  
efivarfs        128K  3.1K  120K   3% /sys/firmware/efi/efivars  
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           455M   0  455M   0% /tmp  
none            1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service  
/dev/nvme0n1p13 989M  96M  826M  11% /boot  
/dev/nvme0n1p15 105M  6.3M  99M   7% /boot/efi  
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service  
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service  
none            1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service  
tmpfs           91M   8.0K  91M   1% /run/user/1000  
/dev/nvme1n1    4.9G  1.3M  4.6G   1% /mnt/mydata  
ubuntu@ip-172-31-33-111:~$ echo "Hello from EBS!" | sudo tee /mnt/mydata/test.txt  
Hello from EBS!  
ubuntu@ip-172-31-33-111:~$ cat /mnt/mydata/test.txt  
Hello from EBS!  
ubuntu@ip-172-31-33-111:~$
```

Screenshot 2026-05-16 021416.png

Linux Commands Used

```
ssh -i C:\Users\Shreyansh\.ssh\Ebs-connect.pem ubuntu@13.233.28.197
lsblk
sudo mkfs -t ext4 /dev/nvme1n1
sudo mount /dev/nvme1n1 /mnt/mydata
df -h
echo "Hello from EBS!" | sudo tee /mnt/mydata/test.txt
cat /mnt/mydata/test.txt
```

Verification Result

The EC2 instance was launched successfully, the EBS volume was created and attached, and Ubuntu detected the 5 GiB disk as /dev/nvme1n1. After formatting the disk with ext4 and mounting it at /mnt/mydata, the test file was written and read successfully. This confirms that the EBS volume is attached, mounted, and usable for storage.

Check	Observed Result	Status
EC2 instance launch	Instance i-0b81e1e5bf5e41eb5 running	Successful
EBS volume creation	5 GiB gp3 volume vol-019e9869a679c8513 created	Successful
Volume attachment	Volume attached to Ebs-connect instance	Successful
Linux detection	lsblk shows /dev/nvme1n1 as 5G disk	Successful
Mount verification	df -h shows /mnt/mydata mounted	Successful
Read/write test	test.txt contains Hello from EBS!	Successful

Conclusion

This practical successfully demonstrates the complete AWS EC2 and EBS workflow: launching an Ubuntu EC2 instance, creating a separate EBS volume, attaching it to the instance, formatting and mounting it in Linux, and validating storage access with a read/write test.